

Model Question Paper

1. $(112 \times 5^4) = ?$
(A) 67000 (B) 70000 (C) 76500 (D) 77200
2. It is being given that $(2^{32} + 1)$ is completely divisible by a whole number. Which of the following numbers is completely divisible by this number?
(A) $(2^{16} + 1)$ (B) $(2^{16} - 1)$ (C) (7×2^{23}) (D) $(2^{96} + 1)$
3. What least number must be added to 1056, so that the sum is completely divisible by 23 ?
(A) 2 (B) 3 (C) 18 (D) 21 (E) None of these
4. $1397 \times 1397 = ?$
(A) 1951609 (B) 1981709 (C) 18362619 (D) 2031719 (E) None of these
5. $(17)^{3.5} \times (17)^? = 17^8$
(A) 2.29 (B) 2.75 (C) 4.25 (D) 4.5
6. Given that $10^{0.48} = x$, $10^{0.70} = y$ and $x^z = y^2$, then the value of z is close to:
(A) 1.45 (B) 1.88 (C) 2.9 (D) 3.7
7. If $5^a = 3125$, then the value of $5^{(a-3)}$ is:
(A) 25 (B) 125 (C) 625 (D) 1625
8. If $3^{(x-y)} = 27$ and $3^{(x+y)} = 243$, then x is equal to:
(A) 0 (B) 2 (C) 4 (D) 6
9. $(256)^{0.16} \times (256)^{0.09} = ?$
(A) 4 (B) 16 (C) 64 (D) 256.25
10. $(0.04)^{-1.5} = ?$
(A) 25 (B) 125 (C) 250 (D) 625
11. $(25)^{7.5} \times (5)^{2.5} \div (125)^{1.5} = 5^?$
(A) 8.5 (B) 13 (C) 16 (D) 17.5
12. Which of the following numbers is divisible by each one of 3, 7, 9 and 11 ?
(A) 639 (B) 2079 (C) 3791 (D) 37911 (E) None of these
13. Which natural number is nearest to 9217, which is completely divisible by 88 ?
(A) 9152 (B) 9240 (C) 9064 (D) 9184 (E) None of these
14. $(4300731) - ? = 2535618$
(A) 1865113 (B) 1775123 (C) 1765113 (D) 1675123 (E) None of these
15. n is a whole number which when divided by 4 gives 3 as remainder. What will be the remainder when $2n$ is divided by 4 ?
(A) 3 (B) 2 (C) 1 (D) 0
16. $397 \times 397 + 104 \times 104 + 2 \times 397 \times 104 = ?$
(A) 250001 (B) 251001 (C) 260101 (D) 261001
17. $(35423 + 7164 + 41720) - (317 \times 89) = ?$
(A) 28213 (B) 84307 (C) 50694 (D) 56094 (E) None of these
18. Which one of the following numbers is completely divisible by 45?
(A) 181560 (B) 331145 (C) 202860 (D) 2033555 (E) None of these
19. Which of the following numbers will completely divide $(3^{25} + 3^{26} + 3^{27} + 3^{28})$?
(A) 11 (B) 16 (C) 25 (D) 30
20. Which of the following statements is not correct?
(A) $\log_{10} 10 = 1$ (B) $\log (2 + 3) = \log (2 \times 3)$
(C) $\log_{10} 1 = 0$ (D) $\log (1 + 2 + 3) = \log 1 + \log 2 + \log 3$
21. If $\log 27 = 1.431$, then the value of $\log 9$ is:
(A) 0.934 (B) 0.945 (C) 0.954 (D) 0.958

22. If $\log_{10} 5 + \log_{10} (5x + 1) = \log_{10} (x + 5) + 1$, then x is equal to:
 (A)1 (B)3 (C)5 (D)10
23. A number when divide by 6 leaves a remainder 3. When the square of the number is divided by 6, the remainder is:
 (A) 0 (B)1 (C)2 (D)3
24. What will be remainder when 17^{200} is divided by 18 ?
 (A)17 (B)16 (C)1 (D)2
25. The difference of the squares of two consecutive even integers is divisible by which of the following integers ?
 (A) 3 (B) 4 (C) 6 (D)7
26. What is the unit digit in $\{(6374)^{1793} \times (625)^{317} \times (341^{491})\}$?
 (A)0 (B)2 (C)3 (D)5
27. The smallest 3 digit prime number is
 (A)101 (B) 103 (C) 109 (D) 113
28. $107 \times 107 + 93 \times 93 = ?$
 (A) 19578 (B) 19418 (C) 20098 (D) 21908
29. How many prime numbers are less than 50 ?
 (A) 16 (B) 15 (C) 14 (D) 18
30. The value of $[(10)^{150} \div (10)^{146}]$
 (A) 1000 (B) 10000 (C) 100000 (D) 10000000
31. Solve for x in the equation $4(x-3)=20$
 (A) 7 (B) 8 (C) 9 (D) 10
32. What is the sum of the interior angles of a triangle?
 (A) 90 degrees (B) 120 degrees (C) 180 degrees (D) 360 degrees
33. Which of the following represents a linear function?
 (A). $y=x^2$ (B) $y=2x+3$ (C) $y=1/x$ (D) $y = \sqrt{x}$
34. What is the domain of the function $f(x)=1/x$?
 (A) All real numbers except 0 (B) All real numbers except 1
 (C) All real numbers (D) Only 0
35. If you are facing North and turn 90 degrees clockwise, which direction are you facing now?
 (A) North (B) South (C) East (D) West

36 Find the value of x: $4x-7=2x+5$

- a)-6
- b)6
- c)-12
- d)12

37Simplify the following exponentiation: $(3^2 \times 3^3) \div 3^4$

- a)243
- b)81
- c)1
- d)3

38A person starts walking from a point. He walks 15 meters towards the North, then turns right and walks 25 meters, then turns right again and walks 15 meters. How far is he from the starting point?

- a) 10 meters
- b) 25 meters

- c) 30 meters
- d) 40 meters

39 The sum of the ages of A and B is 30 years. A is 6 years older than B. What is the present age of A?

- e) 18 years
- f) 20 years
- g) 24 years
- h) 26 years

40 Solve for x if: $\log(x+3)=2$

- i) 97
- j) 80
- k) 100
- l) 99

41 If P is true and Q is false, what is the truth value of the expression $P \wedge Q$?

- m) True
- n) False
- o) Cannot be determined
- p) True only if P and Q are both true

42 The first term of an arithmetic progression is 5,
and the common difference is 3. What is the 10th term of the progression?

- q) 32
- r) 33
- s) 35
- t) 34

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and the common difference is 3. What is the 10th term of the progression?

- u) 32
- v) 33
- w) 35
- x) 34

44 The radius of a circle is 7 cm. What is the circumference of the circle?

- y) 21.98 cm
- z) 43.96 cm
- aa) 35.00 cm
- bb) 44 cm

45 A right-angled triangle has legs of lengths 9 cm and 12 cm.
What is the length of the hypotenuse?

- cc) 13 cm
- dd) 14 cm
- ee) 15 cm
- ff) 16 cm

46 Find the midpoint of the line segment joining the points P(1,2) and Q(3,8).

- a) (2,5)
- b) (4,5)
- c) (2,6)
- d) (3,5)

47 Simplify: $(2^3) \times (2^4)$

- a) 2^7
- b) 2^{12}
- c) 2^6
- d) 2^5

48 Simplify: $\sqrt{5} \times \sqrt{20}$

- a) 10
- b) $5\sqrt{5}$
- c) 20
- d) 5

49A shopkeeper buys a watch for \$500 and sells it at a 20% profit.

How much does he sell the watch for?

- gg) \$300
- b)\$600
- c)\$250
- d)\$500

50 If $an=3n+2$, what is the value of n when $an=11$?

- a)3
- b)5
- c)4
- d)6

51If the time is 3:00 PM, how many degrees will the minute hand be from the hour hand?

- hh) 90 degrees
- ii) 120 degrees
- jj) 75 degrees
- kk) 150 degrees

52A car travels 60 km in 1 hour and 40 minutes.

What is its average speed?

- ll) 36 km/h
- mm) 40 km/h
- nn) 45 km/h
- oo) 60 km/h

53Convert the decimal number 45 to its binary equivalent.

- pp) 101101
- qq) 110101
- rr) 100101
- ss) 111011

54If $\sin\theta=3/5$, Find the value of $\cos\theta$.

- a)4/5
- b)3/4
- c)5/4
- d)2/5

55nWhich of the following numbers is divisible by 7?

- tt) 672
- uu) 729
- vv) 344
- ww) 144

56.Solve for x: $5x+7=2x-4$

- xx) -11
- yy) 11
- zz) -3
- aaa) 3

57. If $\log_2(x) = 5$, then x equals

- a. 16
- b. 32
- c. 64
- d. 128

58. Find the sum of the first 10 terms of the arithmetic progression 3, 7, 11, 15,...

- a. 210
- b. 200
- c. 220
- d. 230

59 Find the sum of the first 10 terms of the arithmetic progression 3, 7, 11, 15,...

- a) 210
- b) 200
- c) 220
- d) 230

60 If $f(x) = x^2 - 3x + 2$, then find $f(4)$.

- a) 10
- b) 12
- c) 14
- d) 16

61 If a quadratic equation has roots 4 and -3, then the equation is

- e) $x^2 - x - 12 = 0$
- f) $x^2 + x - 12 = 0$
- g) $x^2 - x + 12 = 0$
- h) $x^2 + x + 12 = 0$

62 Convert the decimal number 29 to its binary equivalent.

- i) 11101
- j) 10110
- k) 11100
- l) 10011

63 If P is false and Q is true, what is the truth value of $P \vee Q$?

- m) True
- n) False
- o) Cannot be determined
- p) True only if both are true

64 The sum of three consecutive even numbers is 42. Find the largest number.

- q) 16
- r) 14
- s) 12
- t) 18

65 Find the missing number in the series: 2, 5, 27, __, 535, 828

- u) 775
- v) 732
- w) 756
- x) 864

66A right circular cylinder has a radius of 5 cm and height of 10 cm. What is its volume?

- y) $250\pi \text{ cm}^3$
- z) $500\pi \text{ cm}^3$
- aa) $100\pi \text{ cm}^3$
- bb) $125\pi \text{ cm}^3$

67If $(\sin 30^\circ - \cos 60^\circ + \cot^2 45^\circ) / (\cos 30^\circ - \tan 45^\circ + \sin 90^\circ)$, find the value of $\cos \theta$.

- cc) $2/\sqrt{3}$
- dd) $\sqrt{3}/2$
- ee) 1
- ff) $\sqrt{2}$

68Which of the following numbers is divisible by 11?

- gg) 1331
- hh) 1451
- ii) 2461
- jj) 3991

69Find the value of $\cot[-4\pi/3]$

- kk) $1/\sqrt{3}$
- ll) $1/\sqrt{2}$
- mm) 1
- nn) $\sqrt{3}/2$

70A train travels at a speed of 72 km/h. How much distance will it cover in 15 minutes?

- oo) 18 km
- pp) 24 km
- qq) 30 km
- rr) 36 km